INCH-POUND

MIL-E-3677D
27 October 1993
SUPERSEDING
MIL-E-3677C(ME)
7 February 1985

# MILITARY SPECIFICATION

# ERECTION EQUIPMENT

# FOR BRIDGE, FIXED, STEEL,

# PANEL, BAILEY TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers bridge-erection equipment used in erecting a portable panel, Bailey-type bridge.

1.2 Classification. Equipment shall be the following designations, as specified (see 6.2):

Bar, carrying
Extractor, pin
Jack, chord
Lever, panel
Roller, plain
Roller, rocking
Shoe, jack
Template, roller, rocking
Template, roller, plain
Tongs, carrying
Wedge

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| PPP-B-601 | - | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner. |

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-B-121 | - | Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-T-704 | - | Treatment and Painting of Materiel. |
| MIL-G-10924 | - | Grease, Automotive and Artillery. |
| MIL-T-22085 | - | Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-130 | - | Identification Marking of US Military Property. |
| MIL-STD-889 | - | Dissimilar Metals. |
| MIL-STD-1186 | - | Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods. |

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DOCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

| | | |
|---|---|---|
| TA13220E4774 | - | Erection Equipment for Panel Bridge, M2 (Bailey Type). |

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: US ARMY BELVOIR RDE CTR, ATTN SATBE JBS, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).



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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Material.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.1 - Covered Carbon Steel Arc Welding Electrodes.
D 1.1 - Structural Welding Code - Steel.

(Applications for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The erection equipment shall be as shown on Top Assembly TA13220E4774 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the erection equipment. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.



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3.3.1 Material deterioration prevention and control. The erection equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bridge erection equipment may be newly fabricated from recovered materials to the maximum extent practicable, provided the bridge erection equipment produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bridge erection equipment.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A 5.1, classification E6010, E6011, E6012, E6013, or E6020, as prescribed for the operation.

3.5 Initial lubrication service. The erection equipment shall be serviced with standard military lubricants conforming to MIL-G-10924.

3.6 Identification marking. Each item of equipment shall be identified in accordance with MIL-STD-130 and as shown on the drawings.

3.7 Treatment and painting. The erection equipment shall be treated and painted in accordance with MIL-T-704, type A, color forest green. The working parts of the cord jack, pin extractor, roller shafts, and bearings, shall not be painted.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of the erection equipment shall be straight and free from laminations. Straightening of materials shall be done by methods that will not change the physical properties of the materials. All bends shall be made with metal dies or fixtures in order to ensure uniformity of size and shape. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All burrs shall be removed. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are required or where they are not detrimental to safety.

3.8.2 Wood fabrication. All wood parts shall be free from splinters, holes, unsound knots, and other imperfections. The taper of wood parts shall be uniform for the entire length of the taper.

3.8.3 Bolted connections. Boltholes shall be punched or drilled and all burrs shall be removed.



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3.8.4 Rivet and pin connections. Rivet and pin holes shall be drilled to size or punched 1/8 inch under size and reamed to size. Holes shall not have any burrs. Rivet heads shall be concentric with the rivets and shall be in full contact with the surface of the members. The pins or rivets used in rotating or pivoting joints shall allow the joints to operate without binding.

3.8.5 Welders and welding.

3.8.5.1 Welders or welding operators. Welders and welding operators assigned to manual welding work by this specification shall be qualified in accordance with AWS D1.1 or the ASME Boiler and Pressure Vessel code for the materials joined and the type of welding operation to be performed. The contractor is responsible for determining that automatic welding equipment operations are capable of producing quality welds in accordance with AWS and ASME codes (see 6.4).

3.8.5.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded electric-arc method and shall be in accordance with the ASME or AWS code. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and the adjoining base metal shall be wire-brush cleaned. All welded parts shall be assembled in jigs or frames and welded together while held in position. The finished erection equipment shall be within tolerance dimensions. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Straightening and stress relieving shall be permitted to relieve distortion, provided the straightening process does not change the physical characteristics of the metal used in the erection equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.



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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article erection equipment shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for major and minor defects as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The erection equipment shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified on the drawings (see 3.1.1).
102. Material not as specified (see 3.3).
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the bridge erection equipment (see 3.3.2).
106. Components and parts missing or not as specified (see 3.1).
107. Weld defects such as flux inclusions, spatter, undercutting, porosity, or lack of fusion with parent metal (see 3.8.4).
108. Nonfunctioning of parts and components (see 3.8).
109. Edges not having the specified radius; or burrs on metal parts (see 3.8.1).
110. Wood parts containing splinters, knot holes, or unsound knots (see 3.8.2).
111. Workmanship not as specified (see 3.8).
112. Welding and welds not as specified (see 3.8.5).
113. Welders and welding operators not qualified as specified (see 3.8.5).
114. Welding electrodes not as specified (see 3.4).
115. Initial lubrication service not as specified (see 3.5).



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Minor

201. Treatment and painting not as specified (see 3.7).
202. Identification markings incorrect or missing (see 3.6).
203. Screws, nuts, bolts, washers, or cotter pins missing or not as specified (see 3.8.1).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

116. Unfinished or machined surfaces not coated with preservative and wrapped or covered when required, as specified for level A (see 5.1.1).
117. Components of unlike description packed together for level A or B (see 5.2.1 and 5.2.2).
118. Quantities packed together exceed the weight limitation of the box for level A or B (see 5.2.1 and 5.2.2).
119. Blocking, bracing, and anchoring not in accordance with the reference document for level A or B (see 5.2.1 and 5.2.2).
120. Preservation and packing not in accordance with the referenced document as specified for commercial (see 5.1.2 and 5.2.3).
121. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Unpainted unfinished metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. Exposed machine surfaces requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-6 or P-11 preservative and the coated surface covered or wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to MIL-T-22085, type II. The preservatives shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.2 Commercial. Erection equipment shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2).

5.2.1 Level A. Erection equipment of like description shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2 style optional, in quantities not to exceed



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the weight limitation of the box. The contents shall be blocked, braced, and anchored as applicable to prevent free movement in accordance with MIL-STD-1186. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable and strapping shall be zinc-coated.

5.2.3 Commercial. Erection equipment shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129 for levels A and B or ASTM D 3951 for commercial.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The erection equipment is intended for use in erecting a portable panel Bailey-type bridge.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Designation of equipment as desired (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is not required for inspection and approval and the number of units required (see 3.2).
e. Degree of preservation and packing required (see 5.1 and 5.2).
f. Any special marking requirement (see 5.3).

6.3 First article. When a first article inspection is required, the items(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.



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6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference paragraph | DID Number | DID Title |
|---|---|---|
| 3.8.5.1 | DI-MISC-80876 | Welding Procedure Qualification Test Report |
| 3.8.5.1 | DI-MISC-80875 | Welding Procedures |

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.5 Subject term (key word) listing.

Bailey type bridge
Bridge erection equipment
Erection equipment

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activities:
Army - CE
DLA - CT, CS

Preparing activity:
Army - ME

Project 5420-0204



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-E-3677D | 2. DOCUMENT DATE (YYMMDD)<br>931027 |
|---|---|---|

3. DOCUMENT TITLE
Erection Equipment for Bridge, Fixed, Steel, Panel, Bailey Type

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(if applicable)<br>(2) DSN | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. NAME<br>Carolyn B. Johnson | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(703) 704-3468 | (2) DSN<br>654-3468 |
|---|---|---|
| ADDRESS (Include Zip Code)<br>US ARMY BELVOIR RDE CTR<br>ATTN SATBE TSE<br>10101 GRIDLEY RD STE 104<br>FT BELVOIR VA 22060-5818 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>DEFNS QLTY & STDZN OFC<br>5203 LEESBURG PIKE<br>STE 1403<br>FLS CHURCH VA 22041-3466<br>Telephone (703) 756-2340 DSN 289-2340 | |

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